Teacher training and full-command motor distillation

Current proposed full-command architecture: predict missing desired-reference context while preserving the pretrained SONIC motor representation.

(a) Privileged tracking teacher

Repaired G1 motion library



89 train
20 dev

xpriv | rtrue | ht

SONIC actor + critic

release initialization → tracking PPO

Simulator rollouts

tracking rewards + terminations

(b) Public full-command interface — 114 values, 113 available

heading	velocity	height	yaw rate	arrival	keypoints	joints	joint vel.	orientation
2	3	1	1	1	42	29	29	6

Current target only + native measured history $h \in R^{930}$. Arrival coordinate 7 stays masked.

The 79D masked CVAE and 114D transformer are earlier controls; the recovered specialist below is a deterministic reference forecaster.

(c) Full-command inference through the retained motor path

Reference forecaster Fθ

1044 → 512 → 512 → 576 · SiLU

Predicted physical frames

r0 | r̂1 | r̂2 | r̂3 | r̂4 | r̂5 | r̂6 | r̂7 | r̂8 | r̂9

Each frame = qtarget29 + q̇target29 + relative orientation6; pack with the native layout.

h930 + c114 → 9 × 64 residuals

Current frame r0 comes from c.
Future frames: r̂k = r0 + Δrk.
True future appears only in training.

E, Q, D fixed; train Fθ only.
No state-transition forecast is claimed.

Encoder E
640 → 2048 → 1024 → 512 → 512 → 64
FROZEN PRETRAINED

FSQ Q
reshape 2 × 32; 32 levels
FROZEN PRETRAINED

Decoder D
64 tokens + h930 → a29
FROZEN PRETRAINED

(d) Online distillation — teacher actions at the states visited by the student

Query the fixed teacher

Student pre-action state + true reference
→ a*29 and z*64; same-state label
Randomized starts / explicit takeovers

Fresh query + nominal replay

Preserve timing, masks and source identity
Assistance is not autonomous success

Lmotor = MSE(a, a*) + λz MSE(z, z*) + λr Lreference

λr is optional and configuration-bound. Backpropagate through fixed E/Q/D to Fθ.
Then evaluate the public student without future-reference inputs or teacher assistance.

update forecaster → roll out again

Green: action inference. Amber: privileged training inputs. Red dashed: gradient/update path. Gray: frozen components. Reference pose renders are illustrative.

Do not identify this deterministic motor specialist with BFM's CVAE or BFM-Zero's forward–backward representation learning.